Start by opening the source code `chapter_2` folder in VS Code and creating a new Python virtual environment. Again, refer to appendix B if you need assistance.

Then, install the OpenAI and Python dot environment packages using the command in the following listing. This will install the required packages into the virtual environment.

Listing 2.1 `pip` installs

```
pip install openai python-dotenv
```

Next, open the `connecting.py` file in VS Code, and inspect the code shown in listing 2.2. Be sure to set the model's name to an appropriate name—for example, gpt-4. At the time of writing, the `gpt-4-1106-preview` was used to represent GPT-4 Turbo.

Listing 2.2 `connecting.py`

```
import os
from openai import OpenAI
from dotenv import load_dotenv

load_dotenv()                           #1
api_key = os.getenv('OPENAI_API_KEY')
if not api_key:                             #2
    raise ValueError("No API key found. Please check your .env file.")
client = OpenAI(api_key=api_key)                        #3

def ask_chatgpt(user_message):
    response = client.chat.completions.create(      #4
        model="gpt-4-1106-preview",
        messages=[{"role": "system",
 "content": "You are a helpful assistant."},
        {"role": "user", "content": user_message}],
        temperature=0.7,
        )
    return response.choices[0].message.content     #5

user = "What is the capital of France?"
response = ask_chatgpt(user)                #6
print(response)
```

#1 Loads the secrets stored in the .env file
#2 Checks to see whether the key is set
#3 Creates a client with the key
#4 Uses the create function to generate a response
#5 Returns just the content of the response
#6 Executes the request and returns the response